

RESUME

Daeki Hong

Senior Unity/XR Systems Engineer

Unity C# engineer with 10 years of delivery experience across industrial XR, BIM runtime visualization, digital-twin authoring tools, robotics interfaces, and VR simulators. I focus on systems that survive real operating environments: heavy geometry, real-time input, hardware sync, non-developer tooling, installation, debugging, and operator handoff.

Unity C# Runtime

Industrial XR

BIM Geometry

Robotics / IK

Editor Tooling

Field Delivery

CONTACT

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Portfolio darkhtk.github.io/portfolio/

Location Seoul, Korea

BEST FIT

Industrial Unity simulation, XR training, BIM/digital-twin runtime, robotics control interfaces, EditorWindow tooling, and field-delivered software.

EDUCATION

B.S. in Software Engineering, Korea Aerospace University.

I keep this resume focused on work I can explain directly: shipped Unity systems, field tools, ERP workflow screens, and the parts of each project I personally made reliable.

SELECTED WORK

UNITY INDUSTRIAL DELIVERY

VR Robot / DXCenter

Built a Quest 3 - Unity - Flask - RoboDK - Doosan A0912 remote operation flow and a Unity 6 URP/WebGL digital-twin authoring tool, including safety logic, calibration, validation, and non-developer EditorWindow tooling.

HARD ENGINEERING MODULES

IK / BIM / Hardware Sync

Implemented a Jacobian IK solver, runtime BIM section cuts, nine EditorWindow workflows, and hardware-to-Unity synchronization where off-the-shelf tools did not cover delivery constraints.

BIM RUNTIME

WatchBIM / Unity-CrossSection

Delivered custom hit detection, mesh optimization, auto LOD, runtime cross-sections, 2D analysis views, slope labels, minimap, and camera tooling for large road BIM data.

FIELD DELIVERY

VR Simulator Collection

Built construction, factory, naval, FPS safety, and AR training applications with hardware integration, multiplayer mission flow, installation, operation, removal, and operator handoff.

EXPERIENCE

Unity Engineering and Field Delivery

YMX - Senior Engineer, Client Team 2

2024.12 - PRESENT

Industrial Unity development: digital-twin authoring tools, robotics interfaces, and exhibition systems.

- Built a Unity 6 URP/WebGL DXCenter authoring environment with nine EditorWindow tools, ScriptableObject data flow, Service Locator, EventBus, strategy-based state effects, adapter-style device integration, validation flow, and runtime preparation for non-developers.
- Built a Quest 3 - Unity - Flask - RoboDK - Doosan A0912 remote operation pipeline covering coordinate conversion, smoothing, speed/acceleration limits, TCP/HTTP communication, calibration, safety handling, and BEXCO public demo support.
- Implemented an internal Jacobian IK solver to reduce RoboDK dependency, including 6-DOF tracking, damping near singularities, debug UI, and multi-robot validation.

Jangheon Industrial - Seoul Office

2020.06 - 2024.12

BIM visualization, geometry tooling, drawing tools, and engineering software.

- Developed WatchBIM runtime features for highway BIM data, including custom hit engine, mesh optimization, auto LOD, runtime cross-section, 2D analysis viewer, slope label, minimap, and camera tools.
- Built Unity-CrossSection as a reusable C# module using coroutines, NativeArray, Burst-compatible jobs, progress/cancel flow, mesh-plane intersection, caching, and shader visualization.
- Developed road-design drawing tools with WPF/MVVM, MySQL, and CAD entity creation.

Mline Studio / Saltworks / Swink

2015 - 2019

Unity gameplay, VR simulators, Android AR safety training, RS232 device integration, multiplayer mission flow, field installation, operation, removal, and operator handoff.

- Developed construction, factory, and naval safety training VR, FPS safety education, AR education apps, and hardware-linked XR content, then supported customer-site delivery and operator handoff.

UNITY FOCUS

RUNTIME

C# / Unity Runtime

Real-time input, coordinate conversion, state flow, camera control, simulation loops, and operator feedback inside Unity runtime.

TOOLING

EditorWindow Workflow

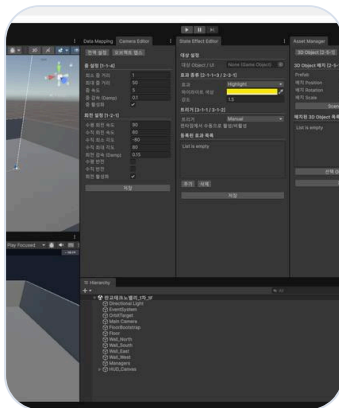
Editor tooling for data mapping, validation, asset registration, and runtime preparation without requiring code changes.

DELIVERY

Hardware / Field

RS232, robot/device links, installation, demos, safety handling, and operator handoff included in the delivery scope.

DXCenter



UNITY 6 URP / WebGL / EDITORWINDOW TOOLING

A digital-twin authoring environment for non-developers. The work packages repeated data mapping, validation, device setup, and runtime preparation into Unity EditorWindow workflows.

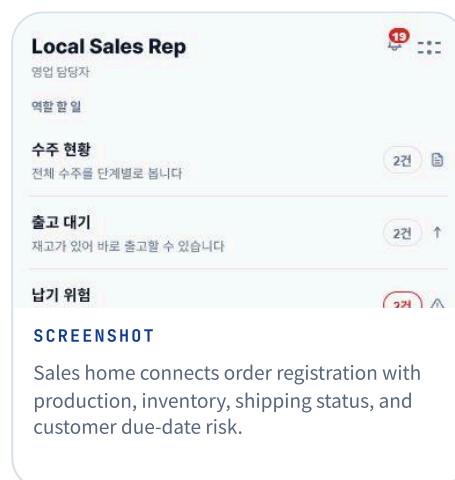
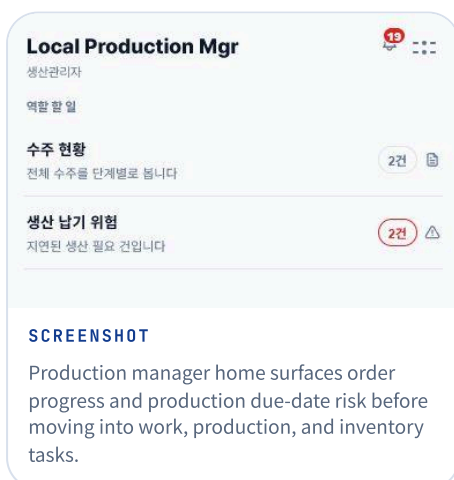
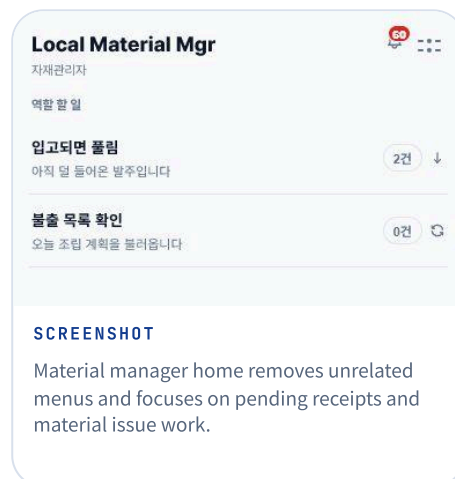
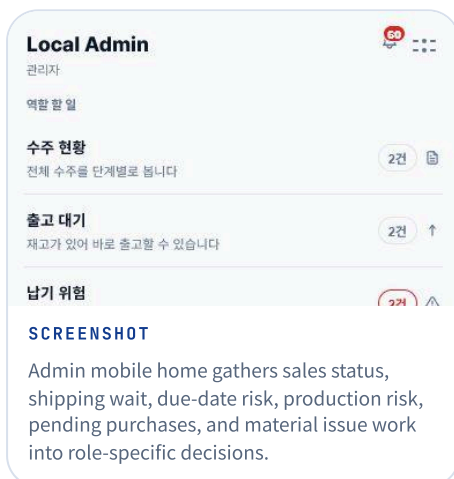
- Designed nine EditorWindow tools around data mapping, validation, asset registration, and runtime readiness.
- Used ScriptableObject data flow, Service Locator, EventBus, strategy-based state effects, and adapter-style device integration.
- I built the editor surface that operators use to prepare content without code.

SEMS ERP

MOBILE-FIRST ERP / FASTIFY TYPESCRIPT / MYSQL / GCP

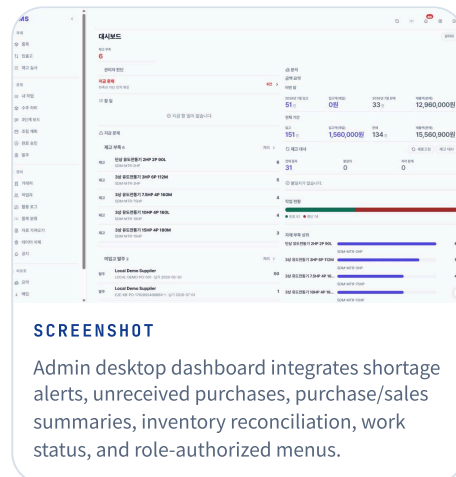
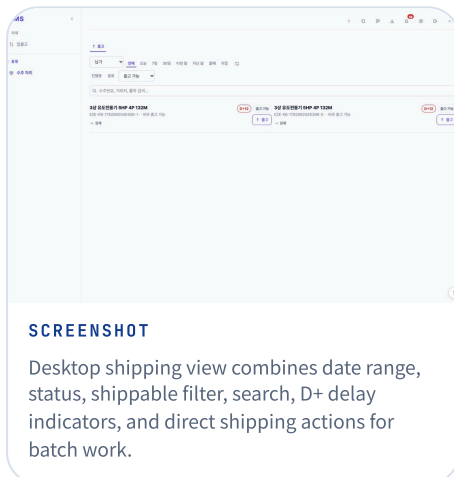
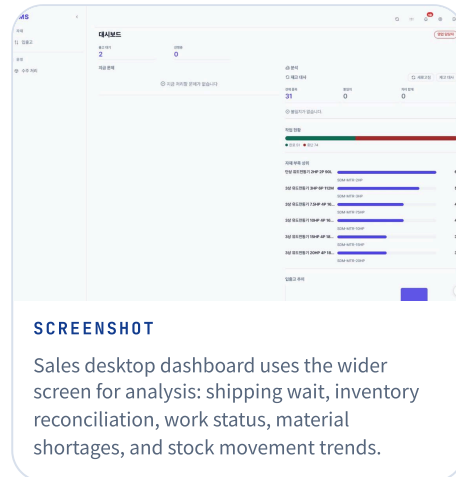
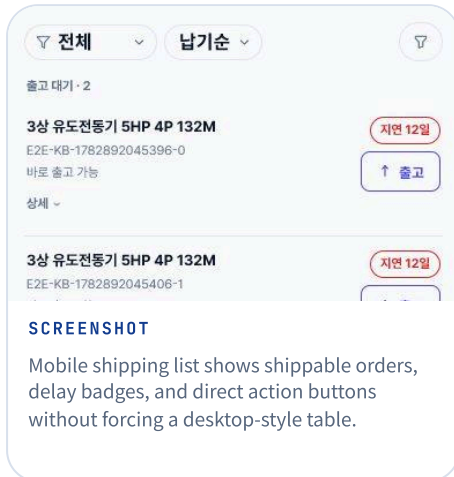
A role-based ERP for small manufacturing teams that connects sales orders, BOM, materials, production, outsourcing, inventory, approvals, shipping, logs, and reporting into one execution flow. The design favors mobile actions over dense desktop tables so field users can complete work from the shop floor.

- Structured manufacturing workflows around roles: admin, material manager, production manager, part leader, shop-floor user, and sales.
- Designed mobile-first task surfaces so each role sees the next action first.
- Implemented the system around Fastify/TypeScript, MySQL/Kysely, Zod validation, Identity Platform auth, Cloud Run, Cloud SQL, Cloud Storage, tests, and mobile/desktop screenshot checks.



SEMS ERP - additional screenshots 1-4 of 4

A role-based ERP for small manufacturing teams that connects sales orders, BOM, materials, production, outsourcing, inventory, approvals, shipping, logs, and reporting into one execution flow. The design favors mobile actions over dense desktop tables so field users can complete work from the shop floor.

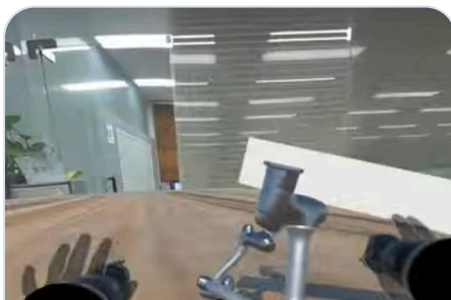


VR Robot Teleoperation



SCREENSHOT

Expo-side view of the remote operation scenario, including the robot and operator environment.



SCREENSHOT

TCP tracking view showing end-effector target following and solver response.

QUEST 3 / UNITY VR / FLASK / ROBODK / DOOSAN A0912

Real-time teleoperation of an industrial robot with a Meta Quest 3. The project combines Unity VR input, safety filtering, a Python bridge, RoboDK integration, and follow-up in-house IK R&D.

- Mapped 6DOF VR controller input into robot target commands with coordinate conversion, smoothing, and safety limits.
- Built the Unity-side feedback loop and bridge assumptions needed for a live BEXCO demonstration.
- Extended the stack with a Jacobian IK solver to reduce external dependency and inspect solver behavior directly.



SCREENSHOT

VR controller input mapped into the IK target so hand motion can drive the robot end-effector.

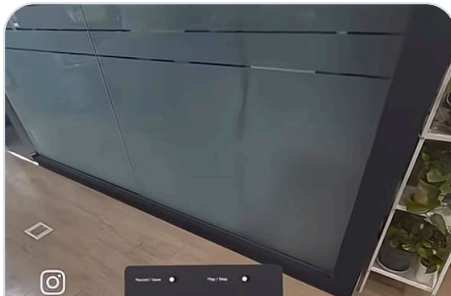


SCREENSHOT

Damping behavior around unstable poses, used to keep the solver usable near singularities.

VR Robot Teleoperation - additional screenshots 1-2 of 2

Real-time teleoperation of an industrial robot with a Meta Quest 3. The project combines Unity VR input, safety filtering, a Python bridge, RoboDK integration, and follow-up in-house IK R&D.



SCREENSHOT

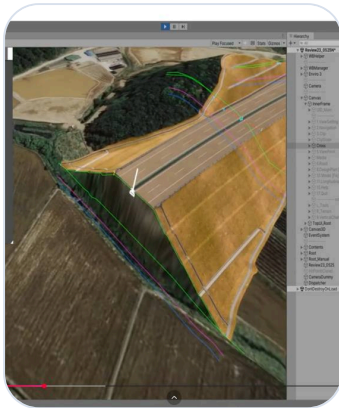
Debug UI for inspecting joints, angles, target error, and solver state during tuning.



SCREENSHOT

Multi-robot validation scene, proving the control model is not hard-coded to a single arm.

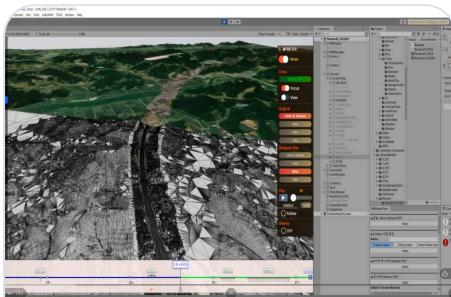
WatchBIM



UNITY BIM RUNTIME / GEOMETRY TOOLING / ANALYSIS VIEWER

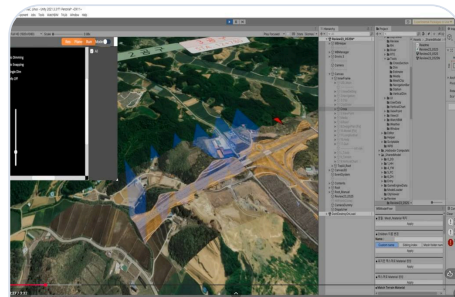
A Unity BIM runtime for large road-model interaction, section cutting, 2D analysis, and camera tooling. The hard part was making heavy geometry usable for operators at runtime.

- Built custom hit detection, mesh optimization, auto LOD, and model interaction flows for large BIM data.
- Implemented runtime cross-sections and linked 2D analysis views for operator checks.
- Added camera/minimap/graph tooling so the BIM model could be navigated and inspected in practice.



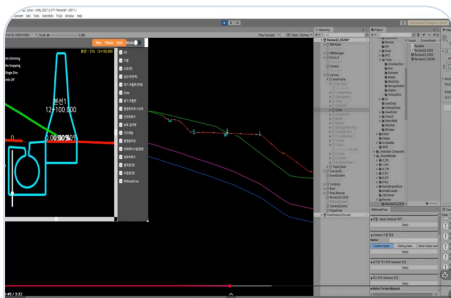
SCREENSHOT

Clipping and selection workflow for navigating large BIM models without relying on external authoring tools.



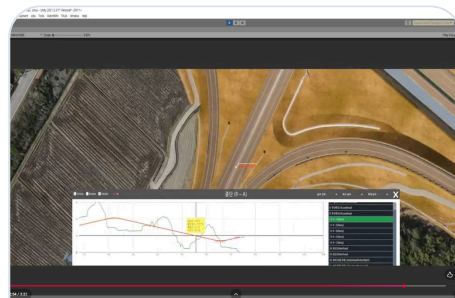
SCREENSHOT

Highlighted section geometry used to make runtime cuts visually clear.



SCREENSHOT

2D analysis viewer linked to the runtime section result for measurement and review.



SCREENSHOT

Vertical graph analysis surface for checking elevation and structural distribution at a glance.

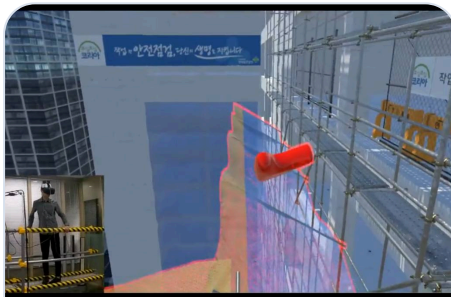
VR Simulator Collection



CONSTRUCTION / FACTORY / NAVAL TRAINING SYSTEMS

Field-delivered VR and AR training projects across construction, factory, and naval scenarios. These projects show installation work, device handling, scripted training flow, and operator handoff.

- Implemented safety training scenes, mission progression, event triggers, and feedback loops in Unity.
- Handled device integration and field constraints such as installation, operation, teardown, and customer handoff.
- Built scenarios where timing, spatial clarity, and training intent mattered more than visual decoration.



SCREENSHOT

Falling accident scenario for construction-site hazard training.



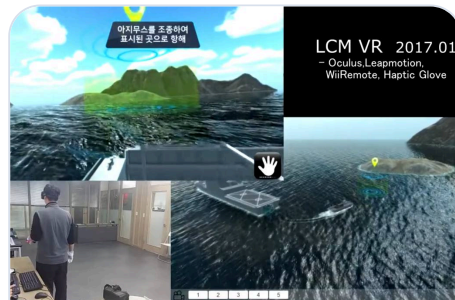
SCREENSHOT

Scaffold scenario focused on spatial awareness and unsafe-work recognition.



SCREENSHOT

Landing craft cockpit simulator view with operator controls and training instrumentation.



SCREENSHOT

Steering interface training for naval simulator operation.

VR Simulator Collection - additional screenshots 1-4 of 4

Field-delivered VR and AR training projects across construction, factory, and naval scenarios. These projects show installation work, device handling, scripted training flow, and operator handoff.



SCREENSHOT

Factory safety training scene for LG Hausys, centered on operator awareness.



SCREENSHOT

Factory accident scenario used to teach hazard response and procedure memory.



SCREENSHOT

Maintenance training context with equipment-state and procedure cues.



SCREENSHOT

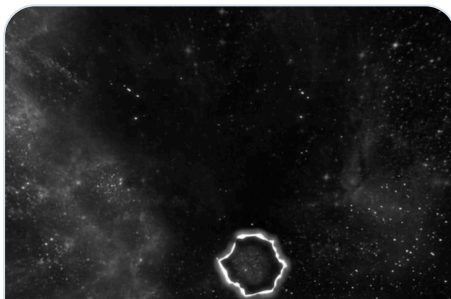
Printer/equipment scenario showing the industrial training environment.

Neostalgia

XR EXHIBITION / VISITOR MEMORY PIPELINE / OFFICIAL SELECTION

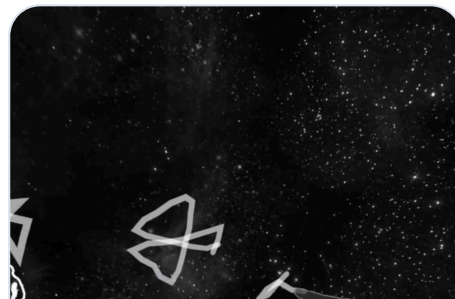
An XR exhibition system selected by BIFAN XR Beyond Reality and recognized with an MSIT citation. It connected visitor input, visual material, and exhibition-grade runtime delivery.

- Built the Unity-side XR experience pipeline and presentation flow for exhibition operation.
- Connected visitor memory, drawing/object assets, dialogue, and skybox presentation into a coherent experience.
- The screenshots show both the generative material and the final immersive runtime surface.



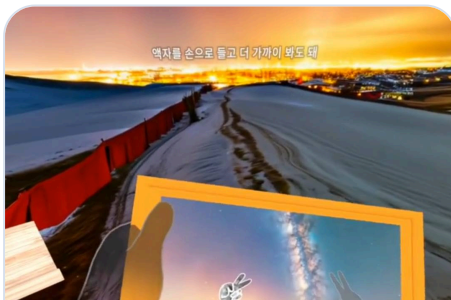
SCREENSHOT

Dialogue/input surface for collecting visitor memory and shaping the exhibition flow.



SCREENSHOT

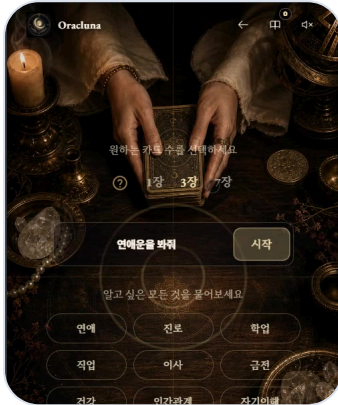
Visitor drawing material used as part of the memory-based exhibition pipeline.



SCREENSHOT

3D object asset placed into the XR scene composition.

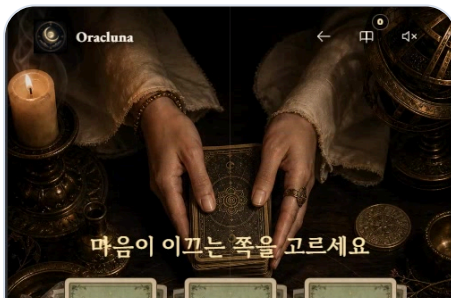
Oracluna Tarot



MOBILE-FIRST WEB APP / 78-CARD UX / QDRANT RAG / DOCKER

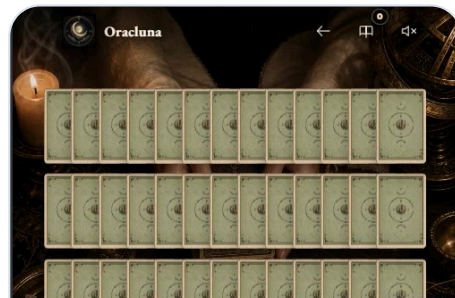
A mobile tarot reading product that carries a user from question, shuffle, and cut through direct selection from all 78 cards, interpretation, saving, and sharing. Each screen keeps one primary action so the cards remain the focus.

- Replaced an unstable radial deck after repeated mobile QA with a full 78-card matrix that keeps every card reachable at a predictable position.
- Built a Qdrant-backed RAG pipeline combining the question, spread position, card, orientation, and retrieved knowledge, with provider adapters for local Ollama development and OpenAI API deployment.
- Automated checks for 78 PNGs, 78 SVGs, 78 metadata files, and 170 retrieval records, then completed iOS Web Audio handling, pre-generated share posters, and Docker/NAS deployment.



SCREENSHOT

Three-way cut after shuffling, preserving the physical rhythm without slowing the mobile flow.



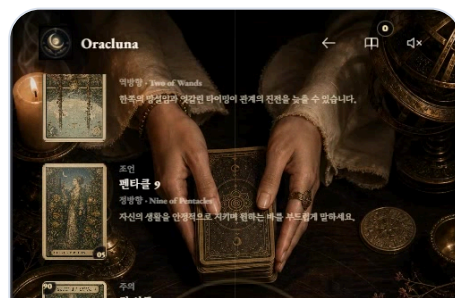
SCREENSHOT

Full 78-card mobile matrix for direct, predictable selection.



SCREENSHOT

Seven-card reveal showing spread positions and upright or reversed orientation.



SCREENSHOT

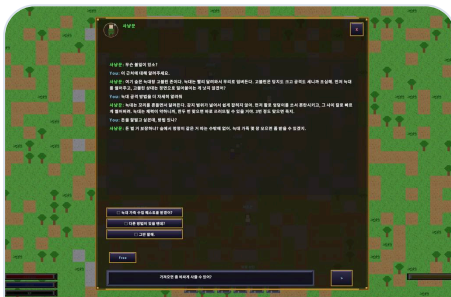
Completed reading with card evidence, saving, and sharing actions.

GenWorld

UNITY 2D / NPC STATE SYSTEM / OFFLINE PLAY

A Unity 2D RPG prototype focused on NPC state, dialogue, skills, and offline-safe behavior. The important engineering point is that the game keeps running even when an optional service is unavailable.

- Designed the dialogue path so the game still works when optional runtime services are unavailable.
- Modeled NPC mood, dialogue, skills, and world state as runtime systems inside Unity.
- Used deterministic rules and runtime state inspection to keep the prototype playable offline.



SCREENSHOT

Dialogue flow where NPC response is driven by local context and runtime rules.



SCREENSHOT

NPC mood/state surface used to inspect behavior changes.



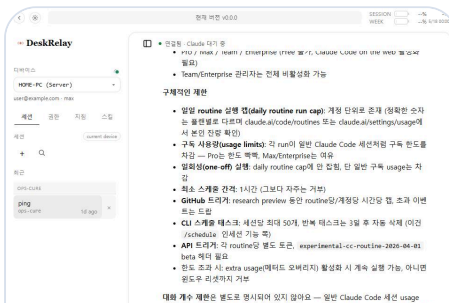
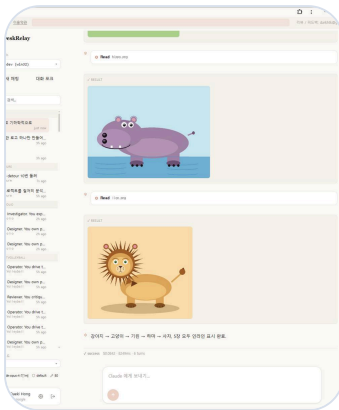
SCREENSHOT

Skill and behavior surface for checking rule-backed NPC actions.

INTERNAL OPERATIONS TOOL / BROWSER AND MOBILE CONTROL SURFACE

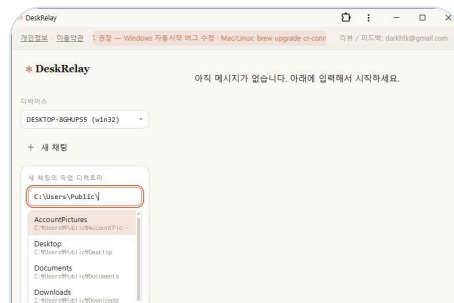
A tool for controlling local work sessions from browser and mobile surfaces. It covers device registration, session continuity, diagnostics, permissions, updates, and mobile UX.

- Designed the control surface around multiple devices, local authority, diagnostics, and repeatable daily usage.
- Built both desktop and mobile flows: chat, workspace selection, device switching, settings, permissions, and update surfaces.
- I worked past the main chat screen into the operational settings that make the tool usable day to day.



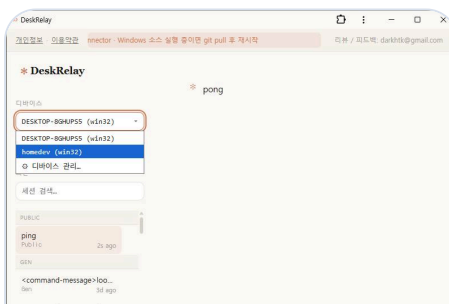
SCREENSHOT

Session screen captured from a working workspace conversation.



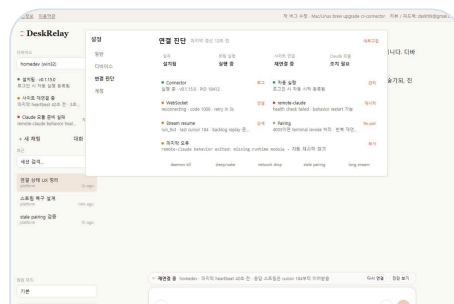
SCREENSHOT

Directory picker flow for choosing a local project safely.



SCREENSHOT

Device switcher showing how one UI can route control to multiple registered machines.

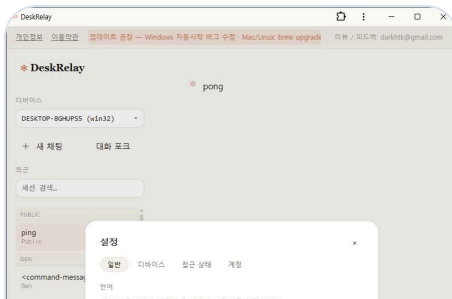


SCREENSHOT

Connection status surface for checking whether the local server and device channel are healthy.

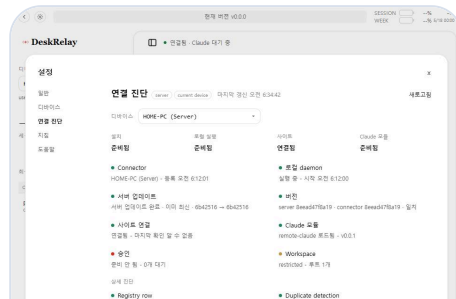
DeskRelay - additional screenshots 1-6 of 8

A tool for controlling local work sessions from browser and mobile surfaces. It covers device registration, session continuity, diagnostics, permissions, updates, and mobile UX.



SCREENSHOT

Settings surface for managing runtime behavior from the product UI.



SCREENSHOT

Diagnostics screen for checking failures without diving into logs first.



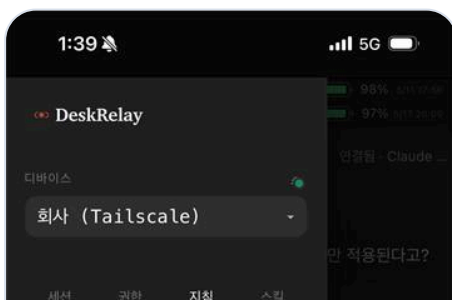
SCREENSHOT

Update settings surface for daily operation and maintenance.



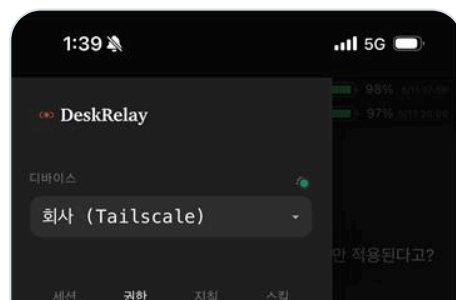
SCREENSHOT

Mobile chat flow for continuing the same coding session away from the desk.



SCREENSHOT

Mobile instructions tab for keeping project guidance visible.

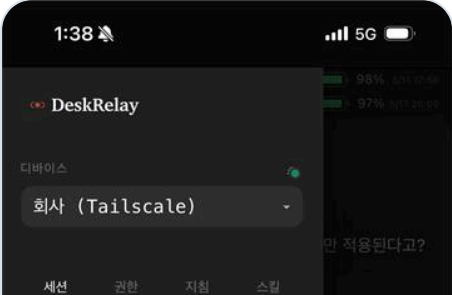


SCREENSHOT

Mobile permissions surface for trust and capability boundaries.

DeskRelay - additional screenshots 7-8 of 8

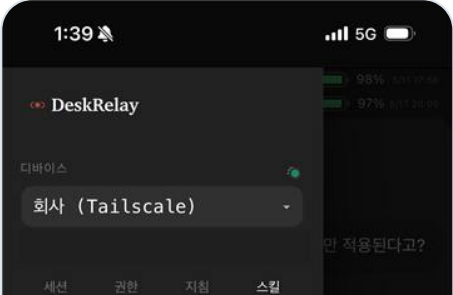
A tool for controlling local work sessions from browser and mobile surfaces. It covers device registration, session continuity, diagnostics, permissions, updates, and mobile UX.



The screenshot shows the DeskRelay mobile app interface. At the top, the status bar displays the time 1:38, signal strength, 5G connectivity, and battery level. The app header includes the DeskRelay logo and a '다바이스' (Device) section with a dropdown menu currently set to '회사 (Tailscale)'. Below this, there is a list of sessions with columns for session name, status, and other details. At the bottom, there is a navigation bar with tabs labeled '세션' (Sessions), '권한' (Permissions), '지침' (Guidelines), and '스킵' (Skip).

SCREENSHOT

Mobile session list for switching work contexts.



The screenshot shows the DeskRelay mobile app interface. At the top, the status bar displays the time 1:39, signal strength, 5G connectivity, and battery level. The app header includes the DeskRelay logo and a '다바이스' (Device) section with a dropdown menu currently set to '회사 (Tailscale)'. Below this, there is a list of available capabilities or skills. At the bottom, there is a navigation bar with tabs labeled '세션' (Sessions), '권한' (Permissions), '지침' (Guidelines), and '스킵' (Skip).

SCREENSHOT

Mobile skills/tooling surface for checking available capabilities.

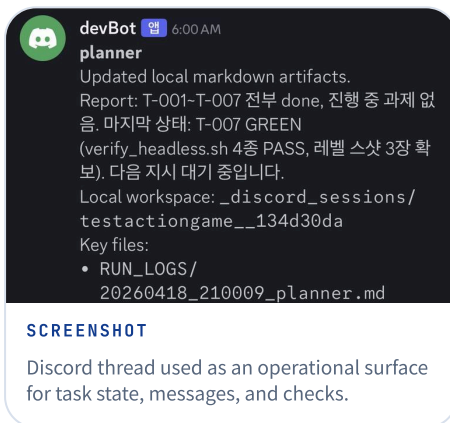
Ops-Cure Workflow



DISCORD OPERATIONS / TASK STATE / WORKFLOW LOOP

A workflow system that keeps planning, task state, execution notes, and handoff visible through Discord threads. I include it briefly because it shows the same habit I bring to tools: make the process inspectable and recoverable.

- Modeled task state so work can move through visible threads without losing context.
- Kept planning, work, checks, and handoff states explicit.
- Used the Discord thread as an operational surface, so state and recovery paths stay readable.



SCREENSHOT

Discord thread used as an operational surface for task state, messages, and checks.